DVCon India 2026 Design Contest

Problem Statement

Genesis of the problem :

Conventional object detection techniques are task-agnostic - they detect all objects in a scene and lack the capability to adapt their detection behavior dynamically based on the task context. Thus, when these techniques are used in situations where objects must be decided based on text semantics, they fall short leading to reduced decision relevance and inefficient computation. Therefore, there is a need for a **task-driven object detection framework** that selectively detects and prioritizes objects **relevant to a given task description or goal**, while ignoring irrelevant objects.

For instance:

Consider the example taken from the reference: “What Object Should I Use? - Task Driven Object Detection” https://doi.org/10.48550/arXiv.1904.03000

Figure 1. What object in the scene would a human choose to serve wine? In the left image, the wine glass is preferred to other drinking glasses. In the right image, neither a wine glass nor other drinking glasses are present. The cup is therefore chosen by the human.

This is easy and very intuitive for humans. However, it is a very challenging task for automated systems or robots.

Problem Statement:

1. Design a **task-aware object selection framework** for object detection. We will be using COCO dataset and 14 Tasks as defined in the reference work.

2. Proposed solution shall be an edge-compatible detection pipeline that uses lightweight language models integrating task input (text prompt) to filter or prioritize detections.

3. Deploy the proposed detection pipeline on an edge device having VEGA Processor, indigenously developed by CDAC Trivandrum at the heart of it. You will be using Genesys-2 FPGA board to deploy the VEGA processor IP along with your novel accelerator that can accelerate the inference performance of your proposed detection pipeline.

4. The input to your system during the inference will be an image from the dataset and the task (from the 14 defined tasks)

5. The output should be the most appropriate object identification in the image as per the given task.

Evaluation criteria:

1. Novelty of the proposed detection pipeline
2. Accuracy and relevance of object detection based on tasks provided as test cases
3. Inference latency
4. CPU and FPGA utilization
5. Power/energy consumption

This is a multi-stage contest and for each stage teams must submit specified deliverables as per the prescribed timelines.

Stage 1:

Teams must complete the following tasks:

1. Familiarize with the COCO dataset and get better understand of the task-aware object detection using literature such as reference work
2. Familiarize with Vega processor system (resources/workshop by CDAC shall be conducted) and Genesys-2 FPGA board
3. Submit a proposal of your novel task-aware object detection pipeline using the given template. Discuss in the detail the rationale behind the proposed framework, its working and how you propose to deploy and accelerate it on the Genesys-2 board.

Deadline for the submission: 10th March 2026